

Scenario 2

1. Scenario: Word Cloud for Popular Keywords

Question: You're given a collection of product reviews. How will you create a word cloud to visualize the most frequent keywords?

Logical Solution:

- Combine all the reviews into a single large text.
- Remove stopwords (like “the”, “and”, “is”) and punctuation.
- Tokenize the text into words and count the frequency of each word.
- Create a word cloud where word size represents frequency — bigger words appear more often in the text.

2. Scenario: Sentiment Analysis for Hotel Reviews

Question: A hotel wants to understand how customers feel about their service. How would you apply sentiment analysis to the reviews?

Logical Solution:

- Preprocess the reviews: lowercase, remove punctuation, tokenize.
- Use a sentiment lexicon or a pre-trained sentiment model to assign polarity scores (positive, negative, neutral).
- Calculate the overall sentiment for each review.
- Provide summaries like: “80% of reviews are positive”, or flag extremely negative ones for further attention.

3. Scenario: Topic Modeling for News Articles

Question: You're given thousands of news articles with no labels. How would you identify the main themes or topics?

Logical Solution:

- Clean and tokenize the text, remove stopwords.
- Use techniques like **LDA (Latent Dirichlet Allocation)** to group words into topics.
- Each article is represented as a mix of topics (e.g., 30% sports, 70% politics).
- From the most frequent words in each topic, assign a topic label manually or automatically.

4. Scenario: Feature Extraction Using CountVectorizer

Question: You want to feed customer messages into a machine learning model. How do you convert the text into numerical features using CountVectorizer?

Logical Solution:

- CountVectorizer turns each word in the corpus into a unique column (feature).
- For each message, it creates a row vector with the count of how often each word appears.
- The result is a sparse matrix that you can use as input to ML algorithms (like logistic regression, naive bayes).

5. Scenario: Choosing Between CountVectorizer and TF-IDF

Question: When analyzing job descriptions, should you use CountVectorizer or TF-IDF Vectorizer? Why?

Logical Solution:

- Use **TF-IDF Vectorizer** when you want to reduce the influence of common words across documents.
- TF-IDF highlights words that are unique and informative for a specific document.
- CountVectorizer would treat all frequent words as equally important — which might not help in identifying unique job-specific terms.

6. Scenario: Using TF-IDF to Find Important Terms in a Resume

Question: You want to extract the most unique and relevant words from a resume. How would TF-IDF help?

Logical Solution:

- Use TF-IDF to weigh each term in the resume: high when it's frequent in that resume but rare in other resumes.
- This helps in identifying specialized skills or rare certifications that make the candidate stand out.
- The top TF-IDF scored words can be used for keyword extraction or matching with job descriptions.